Answers to the questions on Reports 05–11 (2026-11-03)

> **QUESTION**
> What do the numbers and equations in Reports 05–11 mean, and do the claims hold up? Each question from 2026-11-03 is answered here, with small worked examples. Requests that need the model to be rerun are listed at the end of each section; they go into the rebuilt reports.

INPUTS

- Saved outputs of Reports 05–11 (`odor_weighted/`), official-ROI version and old claws
- TNT and TNT3c recordings (`resp_perfly.pkl` , FC z-scores)

OUTPUTS

- reports/QA_2026-11-03_reports_05-11.html, .pdf

SCRIPTS

- scripts/reports/qa_2026_11_03.py (this page)

← Report 12 · All reports

CONTENTS

1 Report 05: the block model
1.1 The sentence about APL, rewritten
1.2 Why the bouton and claw weights of P and L are counted per synapse
1.3 Why the input weights are divided by all of the KC's claw PN synapses
1.4 What each number in "Results of the prediction run" is
1.5 Does similarity rise most for pairs of strong odors? (ms included)
1.6 What goes into the rebuilt Report 05
2 Report 06: the recordings
2.1 What the per-odor effect is, and what it means mechanistically
2.2 Are the per-odor effects real? The statistics
3 Report 07: pinning and the comparison
3.1 Equation 1 with numbers: the per-odor input gain
3.2 Why the ratio is squared: (0.46/0.84)² ≈ 30%
4 Report 08: the inhibition rules
4.1 The four rules as equations
4.2 How the Spearman values were computed, and -aPine's 1.71
4.3 Why odor strength was the mean of TNT and TNT3c, and the TNT-only result
5 Report 09: the noise correction
6 Report 11: the local APL model
6.1 θ in Equation 1
6.2 Equation 3, every variable, with a mock tree
6.3 The C matrix

All numbers below come from the current reports (official ROIs, old 2 µm claws), so they match what you read. The rebuilt reports will use the corrected calyx and the claws you choose, and some values will shift.

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1 Report 05: the block model

1.1 The sentence about APL, rewritten

Old: "APL is one cell whose activity, for each odor, is the average response of the KCs. It inhibits every KC by subtracting a fixed multiple of that activity."

New: "APL is modelled as a single unit with one activity level per odor. Every KC excites APL, so this activity is the mean response of all KCs to that odor. APL then inhibits every KC equally: it subtracts its activity times a gain a from each KC's input. The gain is the same for every odor, but the inhibition is not, because APL's activity is higher for odors that drive more KCs."

1.2 Why the bouton and claw weights of P and L are counted per synapse

In versions P and L, inhibition does not subtract from the KC. It **divides** the input that glomerulus g sends to every KC by $1 + b\,\tilde{p}_g\,A_o$. So $\tilde{p}_g$ has to answer one question: *how strongly is each unit of g's input inhibited?* Raw APL synapse counts answer a different question, because big glomeruli have more of everything. DC1 has 6,125 PN→KC synapses and VA1v 1,965. If both had the same number of APL→PN synapses per bouton, DC1 would still have about 3 times as many in total. The size of a glomerulus's input already enters the model through W_{kg}. Using raw APL counts would count size a second time and make big glomeruli look strongly inhibited simply because they are big. Dividing by the excitatory synapses removes that: p_g is the number of APL synapses that each PN→KC synapse of g "sees".

(In the sentence you copied, the symbol g is typeset as math, which is why it dropped out of the copied text as "glomerulus 's".)

Table 1. Raw counts and the per-synapse weights, hemibrain. Eight hemibrain glomeruli. The raw counts (columns 4 and 7) mostly follow the glomerulus's size (columns 3 and 6). The ratios (columns 5 and 8) are what P and L use, after scaling to a claw-input-weighted mean of 1.

glomerulus	PNs	PN→KC synapses	APL→PN synapses (boutons)	p_g = APL→PN ÷ PN→KC	PN synapses on its claws	APL→KC synapses on its claws	q_g = APL→KC ÷ claw PN synapses
DC1	1	6,125	57	0.009	6,237	372	0.060
DM6	4	5,248	80	0.015	5,324	254	0.048
DM2	2	4,954	100	0.020	4,960	276	0.056
DA1	7	4,687	68	0.015	4,724	357	0.076
DL5	1	2,517	35	0.014	2,557	144	0.056
VM2	2	2,000	13	0.006	2,013	107	0.053
VA1v	3	1,965	30	0.015	2,021	121	0.060
DA2	5	1,800	15	0.008	1,673	103	0.062

Table 2. Why per synapse, in a mock example. A mock example with two glomeruli. Raw counts would rank A as the more inhibited glomerulus; per synapse, B is.

	glomerulus A (big)	glomerulus B (small)	what it means
PN→KC synapses	1,000	200	A sends 5 times more input to KCs
APL→PN synapses on the boutons	100	40	A receives 2.5 times more APL synapses in total
p_g (per PN→KC synapse)	0.10	0.20	but each unit of B's output is inhibited twice as strongly
a KC with half its input from each	$\frac{0.5\,r_A}{1+0.10\,bA_o}$	$\frac{0.5\,r_B}{1+0.20\,bA_o}$	B's half of the input is divided more

q_g (version L) follows the same logic for APL→KC synapses that sit on the claws of glomerulus g: they are divided by the PN synapses on those claws, so a glomerulus with many large claws is not counted as more inhibited only because it has more claw membrane.

1.3 Why the input weights are divided by all of the KC's claw PN synapses

Dividing by the KC's total makes its drive x_{ko} a **weighted average** of its glomeruli's responses, so every KC's drive is on the same scale whatever its size. Without it, the drive would be $\sum_g n_{kg}\, r_{go}$, which grows with the number of synapses. KCs differ a lot in that number: hemibrain KCs have a median of 89 claw PN synapses (5–95% 32–192), and γ KCs have about twice as many as the others (151 on average, against 75 for αβ and 70 for α′β′). With raw counts and one threshold, γ KCs would take most of the responses. The averaging assumes that a KC with more input synapses also needs more input to reach threshold, which is roughly what a larger dendrite (lower input resistance) does.

The denominator also includes claws whose main PN is not in the response table or is multiglomerular. Those claws count as input of 0, so a KC whose claws come mostly from unmeasured glomeruli gets a smaller drive, instead of having its few measured claws scaled up.

This is a modelling choice, and the data do not settle it. The rebuilt Report 05 will add two alternatives so the effect can be seen: raw synapse counts ($W_{kg} = n_{kg}$, one threshold), and equal weight per claw (W_{kg} = share of the KC's claws from g).

1.4 What each number in "Results of the prediction run" is

"r" in Report 05 is always the **Pearson** correlation. Spearman (rank) correlations are written ρ or ρ_s in all reports (for example the rank order in Reports 06 and 09). The rebuilt reports will spell out "Pearson r" or "Spearman ρ" each time.

The r = 0.99 is **not** a correlation between KC responses and ORN responses. "With ORN input" only says which response table drove the model (as opposed to GH146). The correlation is over odors, between how many KCs an odor drives with APL blocked and how many more KCs respond when APL is blocked.

Table 3. Every number in the Results section of Report 05. ϕ_o is the fraction of KCs responding to odor o. All values are for the main setting (20% of KCs respond with APL blocked, 10% intact).

statement	what is computed	over what	value (ORN / GH146)
KCs gained by blocking track odor strength, r = 0.99	Pearson r between $\phi_o^{\text{blocked}} - \phi_o^{\text{intact,U}}$ (KCs gained) and ϕ_o^{blocked} (odor strength); each fraction averaged over the 3 MBs first	16 odors (ORN) / 13 (GH146)	0.991 / 0.995
and so do all versions, r ≥ 0.999	Pearson r between the KCs gained under U and under each other version	odors	0.999 / 0.996 (lowest version)
mean ΔS	mean of $\Delta S_{ab} = S_{ab}^{\text{blocked}} - S_{ab}^{\text{intact,U}}$ (cosine), each pair averaged over the 3 MBs	105 pairs (ORN, ms left out) / 78	+0.076 / +0.041
same pattern in all three MBs	Pearson r between two MBs' ΔS over pairs, for each of the 3 MB pairs; the median of the 3	pairs	0.87 / 0.81
C changes ΔS by 0.029	mean over pairs and MBs of $\lvert\Delta S_{ab}^{C} - \Delta S_{ab}^{U}\rvert$	pairs × 3 MBs	0.029 / 0.019
shuffled counts do much the same	Pearson r between the maps $\Delta S^{C} - \Delta S^{U}$ and $\Delta S^{Cs} - \Delta S^{U}$ (Cs = counts shuffled within class)	pairs (mean over MBs)	0.72 / 0.71

statement	what is computed	over what	value (ORN / GH146)
what is left differs between MBs	as row 4, for $\Delta S^{C} - \Delta S^{Cs}$	pairs	0.34 / 0.20
P vs P0: consistent pattern	as rows 4 and 5, for $\Delta S^{P} - \Delta S^{P0}$	pairs	r 0.87, |diff| 0.034 / r 0.80, |diff| 0.039
M vs M0	mean $\lvert\Delta S^{M} - \Delta S^{M0}\rvert$	pairs × 3 MBs	0.006 / 0.007
L vs P0	mean $\lvert\Delta S^{L} - \Delta S^{P0}\rvert$	pairs × 3 MBs	0.018 / 0.015

1.5 Does similarity rise most for pairs of strong odors? (ms included)

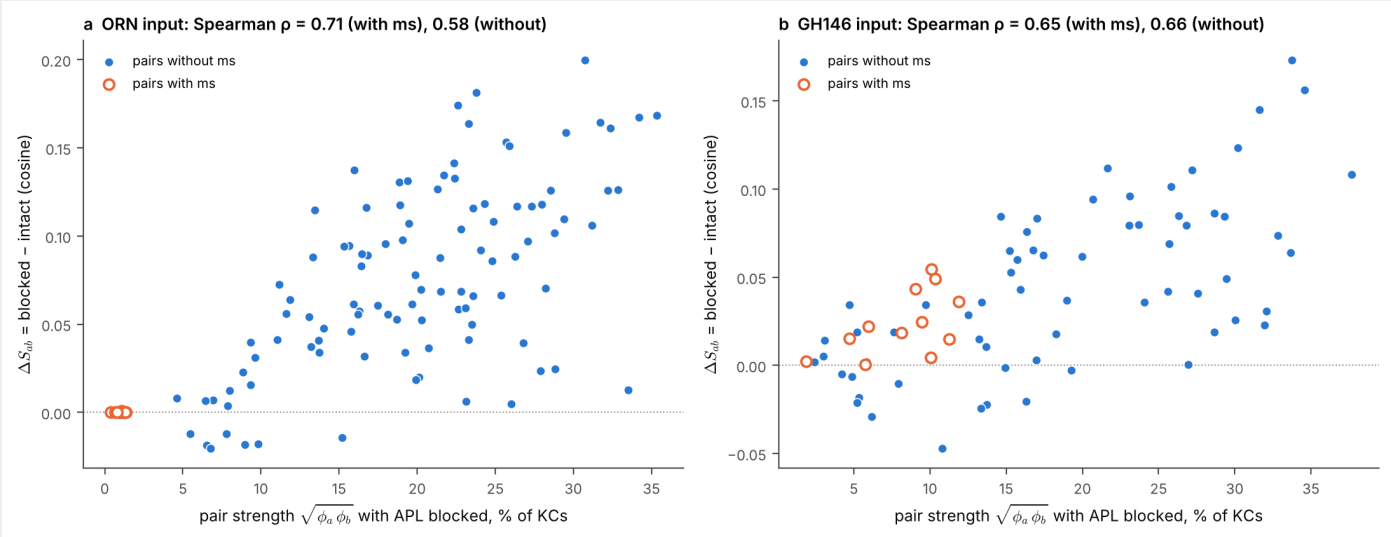


Figure 1. Similarity change against pair strength. Each point is one odor pair (uniform APL, mean of the three mushroom bodies, 20% → 10%). Horizontal: the geometric mean of the two odors' fractions of KCs responding with APL blocked. Vertical: how much more similar the pair is with APL blocked. Open orange circles are the pairs with ms. With ORN input, ms drives 0 KCs in hemibrain R even with APL blocked, so its similarity is undefined there; its pairs here are the mean of male CNS L and R, where it drives 4 and 1 KCs. With GH146 input ms drives 3–4% of KCs.

The claim holds: pairs of strongly driven odors gain the most similarity (Spearman ρ = 0.71 with ORN input, 0.65 with GH146). The pairs with ms sit at ΔS ≈ 0 with ORN input, because ms barely drives any KC: its ORN responses are weak, and with one shared threshold almost no KC reaches it. This is a property of the Report 05 setup only. From Report 07 on, each odor is pinned to its TNT fraction (ms: 11.6% of units), so ms is fully part of every comparison there.

In the rebuilt Report 05, ms stays in every figure and table. Where its similarity is undefined (no KC responds), the cell is marked as such instead of being dropped.

1.6 What goes into the rebuilt Report 05

- **PN-level transformation** (item 1): ORN → PN through the divisive normalization of Olsen et al. (2010), as used in Kennedy's model (2019, "Learning with naturalistic odor representations in a dynamic model of the Drosophila olfactory system"). Its output is compared with the GH146 results. If you meant a different Kennedy paper, tell me.
- **Negative** r_{go} (item 4): a version that keeps inhibitory glomerular responses instead of setting them to 0.

- **Inhibition on both PNs and KCs** (item 5): APL subtracts from KCs and divides bouton input per glomerulus at the same time, with the per-glomerulus bouton weights (like M, with C-style per-KC weights on the KC side).
- **Figures 8 and 9 for every version** (item 8), each caption stating the mean change in similarity with and without inhibition, and a comparison of every version with U.
- **Marked pairs** (item 9): in the ΔS heat maps, the pairs that change most and least are outlined.
- **Data at the end** (item 10): the response-similarity heat maps of the recordings (Report 06, Figure 3), for direct comparison.
- **The two alternative W normalizations** above, and the sentence rewrite.

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2 Report 06: the recordings

Units. The responses are FC z-scores (`fc_zscore`). The rebuilt report says so, and "responding" means a response above 1 z-score unit.

2.1 What the per-odor effect is, and what it means mechanistically

The fit $\Delta S_{ab} \approx \mu + e_a + e_b$ says: the change of a pair is the average change, plus a bit that belongs to each of its two odors. e_a is how much more (or less) the pairs that contain odor a change than the average pair. For example e_{eb} = +0.050: pairs with eb gain about 0.05 more cosine similarity than average when APL is blocked, whatever the partner. An odor with a large e_a has its whole row of the similarity matrix move up.

Mechanism: chance overlap. When an odor drives more units, it shares more responding units with every other odor just by chance (Report 08, Step 2: two unrelated 0/1 codes have cosine $\sqrt{p_a p_b}$). So an odor that gains many responders when APL is blocked should gain similarity with all partners, which is exactly a large e_a. This can be tested on the data alone: compute the chance overlap $\sqrt{p_a p_b}$ from each recording's fractions responding, take its change TNT − TNT3c, fit the same additive model to it, and compare its per-odor effects with the real e_a.

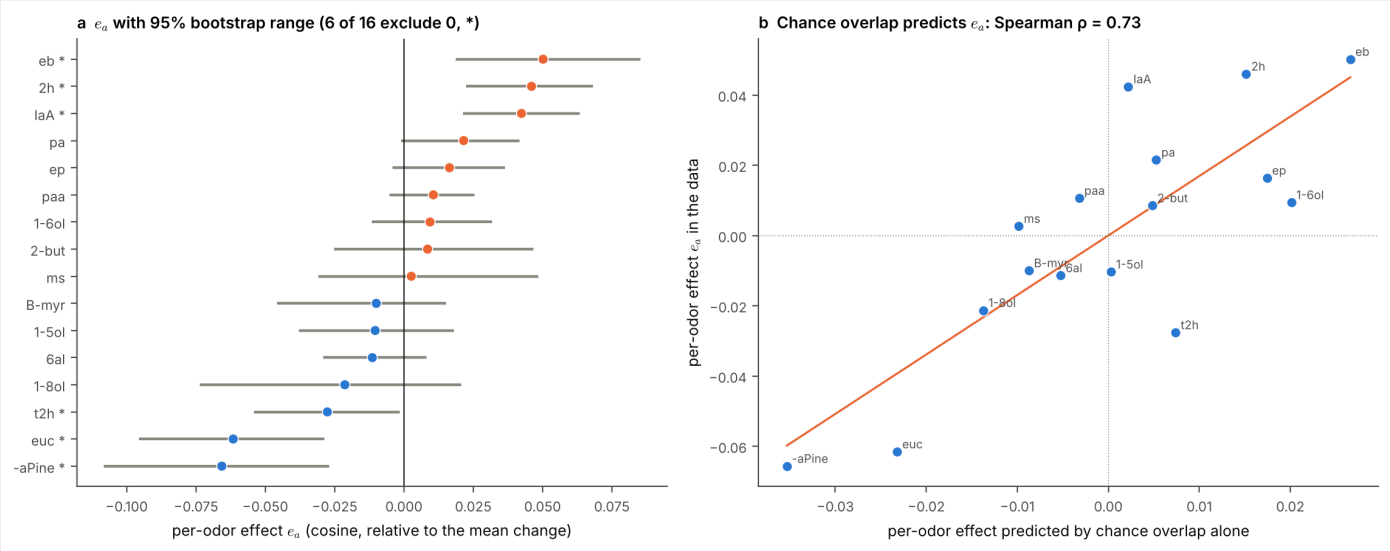


Figure 2. Per-odor effects: how sure, and why. a, The per-odor effects of Report 06, Figure 4a, with their 95% bootstrap ranges (2,000 draws; recordings resampled with replacement within each condition; each draw refits the additive model). **b**, Each point is one odor. Horizontal: the per-odor effect of the additive fit to the change in chance overlap $\sqrt{p_a p_b}$ (fractions responding > 1, per recording, averaged). Vertical: the per-odor effect of the real change in similarity. The orange line is the least-squares fit.

The per-odor effects follow chance overlap (Spearman ρ = 0.73). They also follow how many units an odor drives in TNT (ρ = 0.76) and how many it gains (TNT − TNT3c, ρ = 0.73), but not the ratio TNT ÷ TNT3c (ρ = 0.23). So the per-odor part of the change is mostly the arithmetic of denser codes: odors that drive many units in TNT, such as eb, 2h and IaA, overlap more with everything. -aPine and euc gain the fewest responding units when APL is blocked (2.5 and 4.0 percentage points, against 5.7–13.9 for the other odors) and have the most negative effects.

2.2 Are the per-odor effects real? The statistics

Table 4. Statistics for the per-odor effects. With 16 odors, the additive model has 16 free parameters for 120 pairs, so even random numbers give R^2 ≈ 15/119 ≈ 0.13. That is why the null is needed: 0.65 is far above it.

test	how	result
Is the additive fit better than chance?	Permutation: the 120 values of ΔS are shuffled across pairs (5,000 times), which removes any odor structure, and the additive model is refitted	R^2 = 0.65 in the data; null mean 0.13, 95th percentile 0.20; p = 0.0002 (none of 5,000 shuffles reach it)
Which e_a differ from 0?	95% bootstrap range over recordings (Figure 2a)	6 of 16 odors: -aPine, euc, t2h, IaA, 2h, eb
Are the e_a reproducible?	Split-half: two random halves of the recordings (4 + 5 and 4 + 6), e_a fitted in each, Pearson r over odors; 1,000 splits; Spearman–Brown to full data	r = 0.66 between halves, 0.80 for all the data

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3 Report 07: pinning and the comparison

3.1 Equation 1 with numbers: the per-odor input gain

The goal of pinning is that, with APL blocked, the model drives exactly the fraction of KCs that TNT drives for each odor. The threshold is fixed at θ = 1, and each odor's input is multiplied by a gain g_o chosen so that the right number of KCs end up above 1. Here is the calculation for a mock mushroom body of 10 KCs and one odor that drives 20% of units in TNT:

- **1. Unscaled drives** $\sum_g W_{kg} r_{go}$ of the 10 KCs: 0.12, 0.30, 0.05, 0.42, 0.22, 0.18, 0.60, 0.35, 0.08, 0.50.
- **2. Sort them:** 0.05, 0.08, 0.12, 0.18, 0.22, 0.30, 0.35, 0.42, 0.50, 0.60.
- **3. The quantile.** 20% should respond, so the threshold must sit at the 80th percentile, $Q_o(1 - 0.20) = Q_o(0.80)$. With 10 values its position is 0.80 × 9 = 7.2 places from the bottom (counting from 0): between the 8th value (0.42) and the 9th (0.50), 20% of the way: 0.42 + 0.2 × (0.50 − 0.42) = **0.436**.
- **4. The gain.** $g_o = \theta / Q_o(0.80) = 1/0.436$ = **2.294**.
- **5. Scaled drives** $x_{ko} = g_o \sum_g W_{kg} r_{go}$: 0.28, 0.69, 0.11, 0.96, 0.50, 0.41, 1.38, 0.80, 0.18, 1.15. Exactly two (from 0.60 and 0.50) are above 1, so 2 of 10 = 20% respond, as in TNT. The KC at 0.42 ends just below, at 0.96.

Dividing by the quantile only moves the threshold to the right place in each odor's distribution. The order of the KCs' drives (which KC responds) still comes from the connectome and the ORN responses. An odor with weak ORN responses has small drives, a small quantile, and therefore a large gain.

3.2 Why the ratio is squared: (0.46/0.84)² ≈ 30%

A correlation r is not a share. Its square, r^2, is: the share of the data's variance that the model accounts for. The reasoning goes in three steps.

- **The data are part pattern, part noise.** The split-half reliability of ΔS is 0.702: 70% of its variance over pairs is reproducible pattern, and 30% is recording noise that changes from one set of recordings to the next.
- **A perfect model can explain only the pattern.** Its r^2 would be 0.702, so its r would be $\sqrt{0.702}$ = 0.838. That is the ceiling of 0.84.
- **The uniform model.** r = 0.46, so r^2 = 0.207: it accounts for 21% of all the variance, which is 29% of the 70% that is pattern. That is $r^2/\text{reliability} = (r/\text{ceiling})^2$.

A quick simulation confirms it: 120 mock pairs whose "data" are 70% pattern and 30% noise, and a mock model that shares 30% of its variance with the pattern. The model correlates with the noisy data at r = 0.45 on average and with the noise-free pattern at r = 0.54 (0.55^2 = 0.30). The unsquared ratio 0.46/0.84 = 0.55 is that second correlation; squaring turns it into a share.

Figure 4 will get a $y = x$ line in the rebuild (both axes are blocked ÷ intact ratios, so points on the line mean the model gets that odor's suppression right).

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4 Report 08: the inhibition rules

4.1 The four rules as equations

All four rules start from the same pinned, blocked responses $y_{ko}^{\text{blocked}} = [x_{ko} - 1]_+$ and all give the same mean fraction responding over odors (TNT3c's, 6.83%). They differ only in how the inhibition is shared out between odors. Your reading of the fixed rule is right.

Table 5. The inhibition rules of Report 08. c and c_o are found by bisection in each mushroom body. In all four, the inhibition is the same for every KC for a given odor.

rule	intact response	how the inhibition is set	is it a model?
feedback (version U)	$[x_{ko} - 1 - a\,A_o]_+$, with $A_o = \frac{1}{N}\sum_k y_{ko}$	one gain a for all odors; A_o comes out of the KC responses themselves, so strong odors get more inhibition	yes: the model of Reports 05 and 07
fixed	$[x_{ko} - 1 - c]_+$	one number c for all KCs and all odors (per mushroom body): the same as raising the threshold from 1 to $1 + c$	no: a reference rule
same share	$[x_{ko} - 1 - c_o]_+$	one c_o per odor, chosen so that every odor keeps the same share of its blocked responders: $\phi_o^{\text{intact}} = \phi_o^{\text{blocked}} \times \bar{\phi}^{\text{TNT3c}}/\bar{\phi}^{\text{TNT}}$ (about 45%)	no: a reference rule
TNT3c per odor	$[x_{ko} - 1 - c_o]_+$	one c_o per odor, chosen so that $\phi_o^{\text{intact}} = \phi_o^{\text{TNT3c}}$, the data's own fraction	no: uses the answer from the data

The three extra rules are not new models of APL. They are controls that keep everything the same except how the inhibition depends on the odor: not at all (fixed), in proportion to the odor's own blocked response (same share), or exactly as in the data (TNT3c per odor). Comparing feedback with them shows that the odor-dependence created by feedback is what matters for the fit.

4.2 How the Spearman values were computed, and -aPine's 1.71

For each of the 16 odors, two numbers, each the mean over the three mushroom bodies: the fraction responding with APL blocked (ϕ_o^{blocked}, which equals the TNT fraction by pinning) and the gain $\gamma_o = \phi_o^{\text{blocked}} / \phi_o^{\text{intact}}$. The Spearman correlation over the 16 odors between them is +0.96 for feedback and −0.62 for the fixed rule.

For -aPine, ϕ^{blocked} = 6.07% and ϕ^{intact} with feedback = 3.55%, so γ = 6.07 / 3.55 = **1.71**, the smallest of the 16 (Report 08, Figure 1b, leftmost blue and orange points). In Figure 1a its blue dot is hidden under the orange diamond of the data, whose value is almost the same (1.70). Two things to fix in the rebuild: the dots will be drawn so none hides another, and γ_o will be called the "blocked ÷ intact ratio", because "gain" is also used for the input gain g_o of Report 07. For -aPine that one is large, since its ORN responses are weak.

4.3 Why odor strength was the mean of TNT and TNT3c, and the TNT-only result

I used the mean of the two conditions as a neutral measure of how strongly an odor drives the units, one that does not lean towards either condition. You are right that TNT alone is the better choice: TNT has no APL, so it measures the feed-forward drive directly, and it is also what the model's horizontal axis uses. TNT3c is already flattened by feedback, so mixing it in blurs the measure. The strength and the gain still come from separate halves of the recordings (4 TNT recordings for strength; the other 4 TNT and 6 of the TNT3c recordings for the gain), so they share no noise.

Table 6. The feedback signature with TNT-only odor strength. Strength and gain use the same measure in each row (here, unlike Report 08, where strength was always the fraction > 1). The model with uniform feedback predicts ρ = +0.96 for the fraction.

odor strength measured as	gain = ratio of	mean Spearman ρ (2,000 splits)	5–95%	share of splits > 0
mean of TNT and TNT3c (Report 08)	fraction > 1	+0.06	-0.22 to +0.33	65%
TNT only	fraction > 1	+0.13	-0.13 to +0.41	79%
TNT3c only	fraction > 1	−0.01	-0.30 to +0.26	49%
mean of TNT and TNT3c (Report 08)	mean positive response	+0.27	+0.04 to +0.52	97%
TNT only	mean positive response	+0.34	+0.11 to +0.61	100%
TNT3c only	mean positive response	+0.21	-0.04 to +0.46	91%

With TNT alone the signature is a bit stronger but the conclusion does not change: for the number of responding units it is weak (ρ = +0.13, positive in 79% of splits), far from the +0.96 that uniform feedback predicts; for response size it is clear (ρ = +0.34, positive in 100% of splits). The rebuilt Report 08 will use TNT-only strength.

5 Report 09: the noise correction

Equation 3 is Spearman's correction for attenuation. The logic, step by step:

- **1. Every measurement is pattern + noise.** The similarity of a pair measured from a set of recordings is its true value in that condition plus recording noise. A different set of recordings gives the same true value with different noise.
- **2. Noise lowers correlations.** If two measurements X and Y each carry independent noise, their correlation is the correlation of their true values times a shrink factor: $\mathrm{corr}(X, Y) = \mathrm{corr}(T_X, T_Y) \cdot \sqrt{\mathrm{rel}_X\,\mathrm{rel}_Y}$, where rel = (variance of the true values) ÷ (variance of the measurement) is the share of a measurement that is not noise.
- **3. The within-condition correlation measures rel.** Two halves of the same condition share the true values and differ only in noise, so their correlation equals rel: $\rho^{\mathrm{TNT}}_{\mathrm{within}} \approx \mathrm{rel}_{\mathrm{TNT}}$, and the same for TNT3c.
- **4. Undo the shrink.** ρ_{between} (a TNT half against a TNT3c half) is the true TNT-vs-TNT3c correlation shrunk by $\sqrt{\mathrm{rel}_{\mathrm{TNT}}\,\mathrm{rel}_{\mathrm{TNT3c}}}$. Dividing by that factor gives Equation 3.
- **5. Why halves on both sides.** The between value is computed from half-size samples, exactly like the within values, so both carry the same amount of noise and no Spearman–Brown step is needed.

Table 7. The correction in a simulation. Mock data: 120 pairs, a true TNT3c pattern and a true TNT pattern that correlate as in column 1, and noise in each half sized so that two halves of one condition correlate at about 0.90, as in the recordings. Means of 2,000 simulations. In the first row blocking APL keeps the order perfectly, yet the between value is only about 0.90 because of noise; the correction brings it back near 1. In the second row the between value drops to about 0.83, as in the data, and the correction recovers the true value.

true correlation (Pearson)	true ρ (ranks)	$\rho^{\mathrm{TNT3c}}_{\mathrm{within}}$	$\rho^{\mathrm{TNT}}_{\mathrm{within}}$	ρ_{between}	corrected
1.00	1.000	0.905	0.905	0.905	**1.000**
0.92	0.908	0.905	0.905	0.826	**0.913**
data	?	0.904	0.906	0.831	**0.92**

So a between value of 0.83 with within values of 0.90 is what a true rank correlation of about 0.92 looks like through this much noise. The correction is exact for Pearson correlations with independent noise and a close approximation for Spearman, as the table shows.

6 Report 11: the local APL model

6.1 θ in Equation 1

Yes, θ = 1. Because each odor's input is pinned (Report 07), the threshold can be any positive number: the gains g_o scale the inputs to it. Setting θ = 1 just fixes the units, so x_{ko} is measured in "threshold units" (1 = exactly at threshold).

6.2 Equation 3, every variable, with a mock tree

What it computes. For the local model we need Φ_{vj}: the total signal that KC j's inputs to APL produce at a point v of APL, each input decayed by its distance along APL, $\Phi_{vj} = \sum_{t \in \mathrm{in}(j)} e^{-d(v,t)/\lambda}$. Summing it over

KC k's output sites gives M_{kj}. Equation 3 computes Φ for every node without ever listing distances, using one fact: the decay along a path is the product of the decays of its edges, $e^{-(\ell_1+\ell_2)/\lambda} = e^{-\ell_1/\lambda}\, e^{-\ell_2/\lambda}$.

Table 8. Variables of Equation 3.

Symbol	What it stands for	Values, units, range
$v,\ w$	Nodes of the (reduced) APL skeleton tree. One node is picked as the root; every other node has one parent and may have children	nodes
$\mathrm{parent}(v),\ \mathrm{children}(v)$	The neighbour of v one step towards the root, and the neighbours one step away from it	
ℓ_v	Length of the edge from v up to its parent	µm
λ	Length constant: the distance over which a passive signal falls to 37%	µm
$\kappa_v = e^{-\ell_v/\lambda}$	The share of a signal that survives crossing the edge from v to its parent (either way)	0 to 1
n_{vj}	Number of KC j's KC→APL synapses at node v: where its signal enters APL	count
up_{vj}	Signal at v from KC j's inputs **in v's own subtree** (v and everything below it), each decayed by its distance to v	≥ 0
Φ_{vj}	Signal at v from **all** of KC j's inputs, anywhere on the tree	≥ 0
$M_{kj} = c\sum_{s\in\mathrm{out}(k)} \Phi_{sj}$	Coupling: KC j's signal summed over the nodes where APL synapses onto KC k	

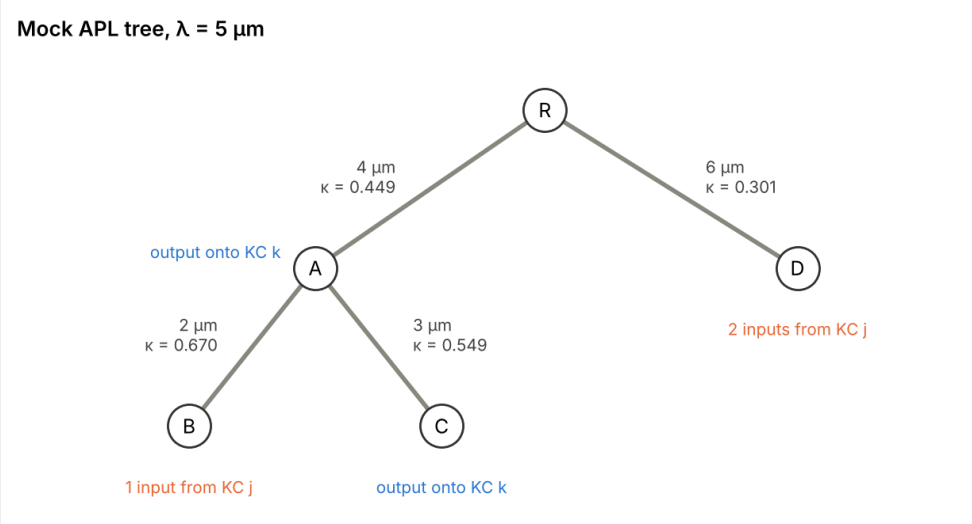


Figure 3. A mock tree for Equation 3. Five nodes. KC j excites APL with one synapse at B and two at D (orange). APL inhibits KC k at A and C (blue). Edge lengths and their decay factors κ are written on the edges.

Pass 1, leaves to root ($\mathrm{up}_{vj} = n_{vj} + \sum_w \kappa_w\, \mathrm{up}_{wj}$): each node collects what enters below it, decayed across each edge on the way up.

Pass 2, root to leaves ($\Phi_{vj} = \mathrm{up}_{vj}(1-\kappa_v^2) + \kappa_v\, \Phi_{\mathrm{parent}(v),j}$). The parent knows the total signal, Φ_{parent}. Bringing it down to v costs one more crossing of the edge, so it is multiplied by κ_v. But part of Φ_{parent} came from v's own subtree: that part reached the parent as $\kappa_v\, \mathrm{up}_v$, and bringing it back would multiply it

by κ_v again, giving $\kappa_v^2\,\mathrm{up}_v$ although its true value at v is up_v. So: take $\kappa_v\Phi_{\mathrm{parent}}$, remove the wrong $\kappa_v^2\,\mathrm{up}_v$ and add the right up_v. That is $\Phi_v = \kappa_v\Phi_{\mathrm{parent}} - \kappa_v^2\,\mathrm{up}_v + \mathrm{up}_v$, which is Equation 3.

Table 9. Equation 3 on the mock tree. Pass 1 is filled from the bottom row up (B, C, D, then A, then R); pass 2 from R down. The last column sums over the three input synapses directly, using the distances along the tree; it equals pass 2 at every node.

node v	parent	ℓ_v (µm)	κ_v	n_{vj}	pass 1: up_{vj}	pass 2: Φ_{vj}	check: $\sum_t e^{-d(v,t)/\lambda}$ directly
R (root)	–	–	–	0	0 + 0.449 × 0.670 + 0.301 × 2.000 = 0.904	= up = 0.904	0.904
A	R	4	0.449	0	0 + 0.670 × 1.000 + 0.549 × 0.000 = 0.670	0.670 × (1 − 0.449²) + 0.449 × 0.904 = 0.941	0.941
B	A	2	0.670	1	1	1.000 × (1 − 0.670²) + 0.670 × 0.941 = 1.181	1.181
C	A	3	0.549	0	0	0.000 × (1 − 0.549²) + 0.549 × 0.941 = 0.516	0.516
D	R	6	0.301	2	2	2.000 × (1 − 0.301²) + 0.301 × 0.904 = 2.091	2.091

For example at A: the input at B is 2 µm away (decay 0.670), and the two inputs at D are 4 + 6 = 10 µm away (0.135 each), so Φ_A = 0.670 + 2 × 0.135 = 0.941. KC k receives APL output at A and C, so $M_{kj} = c\,(\Phi_A + \Phi_C) = c\times$ 1.457. On the real skeletons the same two passes run for every KC j at once.

6.3 The C matrix

Yes, it is the same per-KC weighting as version C of Reports 05 and 07, written as a matrix: $M_{kj} = u_k\,v_j/N$, with u_k = KC k's APL→KC synapses and v_j = KC j's KC→APL synapses in the calyx, each divided by its mean. Putting it into Equation 1 gives $a\sum_j M_{kj}\,y_{jo} = a\,u_k\cdot\frac{1}{N}\sum_j v_j\,y_{jo} = a\,u_k\,A_o$, which is exactly version C: one APL activity A_o (the v-weighted mean KC response), applied to KC k in proportion to u_k. M is "C" only to compare it with the local matrices, which have the same form but let the coupling depend on where on APL the synapses are.